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Unfolding the Network Structure of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ Ionic Liquid Using the BCl_4^- Probe

Gioele Pagot, Mounesha N. Garaga, Keti Vezzù, Boris Itin, Steven G. Greenbaum,* and Vito Di Noto**

Magnesium batteries are emerging as a cutting-edge energy storage solution poised to revolutionize the battery industry. In this context, electrolytes play a pivotal role in the development of reliable and efficient devices. A family of new ionic liquid-based electrolytes is proposed for potential application in magnesium batteries. These electrolytes are synthesized through the reaction of 1-butyl-1-methylpyrrolidinium chloride ($Pyr_{14}Cl$), boron trichloride, and varying amounts of δ-$MgCl_2$. The thermal, structural, and coordination properties of the proposed electrolytes are studied using density functional theory calculations, high resolution thermogravimetry, modulated differential scanning calorimetry, and Fourier transform infrared investigations. For the first time, the structure and nature of interactions characterizing the anionic and cationic domains of ionic liquid electrolytes, as a function of δ-$MgCl_2$ doping, are examined using BCl_4^- as a spectroscopic "ion probe". These results are combined with features of electric response, as well as structural and local relaxation events investigated through broadband electrical spectroscopy and nuclear magnetic resonance studies, respectively, to elucidate the unique characteristics and conductivity mechanisms of the proposed materials. Finally, the reasonably high ionic conductivity of these electrolytes at room temperature makes these materials very promising for application in magnesium battery prototypes.

1. Introduction

The quest for advanced energy storage technologies has become increasingly imperative with the growing demand for portable electronics, electric vehicles, and renewable energy systems.[1] Rechargeable magnesium batteries have emerged as a promising contender in the pursuit of high-energy-density and environmentally sustainable energy storage solutions.[2] Magnesium, an abundant and lightweight metal, offers several unique advantages over traditional lithium-ion batteries, including higher energy density and inherent safety features.[3] Indeed, one of the most significant benefits of magnesium batteries is their inherent safety characteristics. Unlike lithium, which can be prone to thermal runaway and combustion under certain conditions, magnesium exhibits a higher tolerance to abuse, making magnesium batteries less prone to hazardous situations. This improved safety profile is especially crucial for applications where safety is paramount, such as aerospace and portable electronics.

While rechargeable magnesium batteries offer a range of advantages, they also face certain challenges and drawbacks. One of the primary challenges in developing magnesium batteries is finding suitable electrolytes. Traditional electrolytes used in lithium-ion batteries are often not compatible with magnesium, as magnesium ions have a different charge, size, and chemistry, leading to limited ion mobility and efficiency.[4] The electrolyte in a magnesium battery plays a pivotal role in facilitating the reversible electrochemical reactions between the anode and cathode. The selection of appropriate electrolytes is crucial to overcoming

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the challenges associated with magnesium battery technology.[5] Ionic liquids (IL), with their unique properties, have emerged as potential candidates for electrolytes in magnesium batteries.[6] These molecular salts are in a liquid state and exhibit high ionic conductivity, thermal stability, and wide electrochemical stability windows. Their use in magnesium batteries has the potential to address the compatibility issues and improve the overall electrochemical performance of these systems.

The understanding of ion interactions and of the conduction mechanisms occurring in the electrolyte is a critical step in the advancement of magnesium battery technology.[7] Despite promising progress in developing high-conductivity IL,[5d,8] organic ionic plastic crystal,[9] and quasisolid-state electrolytes[10] for magnesium batteries, the mechanisms governing Mg^{2+} solvation, coordination geometry, and ion mobility remain incompletely understood. In particular, a gap exists in probing how multivalent ions interact with structured IL matrices and how these interactions influence mesoscale organization and transport properties. In this report, we summarize results of a study on pyrrolidinium-based IL for applications as potential electrolytes in magnesium secondary batteries. These innovative materials are obtained by reacting first 1-butyl-1-methylpyrrolidinium chloride ($Pyr_{14}Cl$) with boron trichloride to obtain the IL. In a second step, different amounts of δ-$MgCl_2$ are added to IL, yielding eight electrolytes at various magnesium salt concentrations with the formula $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$, where $0 \leq x \leq 0.095$ (x is the molar ratio of Mg to IL unit formula, see **Table 1**).

In detail, the relevant chemical reactivity as a strong Lewis acid makes δ-$MgCl_2$ a highly soluble salt in organic solvents with respect to the "α" phase (i.e., $MgCl_2$ in the ordinary crystalline phase). This is the result of the Mg—Cl covalent bonds of δ-$MgCl_2$, which along the backbone axis form planar polymeric chains ($[MgCl_2]_n$) with square planar $MgCl_4$ repeat units.[6a] The boron trichloride molecule is selected owing to its high reactivity toward free chloride ions as a Lewis acid site, forming BCl_4^- species. This latter is: 1) an excellent ligand for δ-$MgCl_2$, giving rise to highly stable $[BCl_4MgCl_2]^-$ dimer building blocks; and 2) an efficient "spectroscopic structural probe" for studying the structure and the interactions of ILs nanodomains. To the best of our knowledge, this is the first study that systematically employs $[BCl_4]^-$ as a vibrational and structural reporter to track the coordination environment and dynamic reorganization of ionic nanodomains in magnesium-based IL electrolytes. This original approach allows correlating macroscopic transport phenomena with the molecular-scale evolution of the anionic network. In addition, the rational design of $Pyr_{14}Cl$, BCl_3, and δ-$MgCl_2$ components is expected to lead to highly dynamic ionic networks, fostering efficient Mg^{2+} transport and highlighting their potential as next-generation electrolytes.

Table 1. Composition of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes.

Sample	δ-$MgCl_2$ [wt%]	$x = n_{Mg}/n_{IL}$ [a)]	$y = n_{Mg}/n_B$ [a)]	Concentration region
IL	0	0	0	A
1	0.56	0.012	0.048	A
2	0.95	0.020	0.080	A
3	1.46	0.032	0.128	B
4	1.87	0.041	0.164	B
5	2.52	0.056	0.224	B
6	3.15	0.071	0.284	C
7	3.99	0.095	0.380	C

a) n_{Mg}, n_{IL} and n_B are the moles of δ-$MgCl_2$, $[Pyr_{14}Cl/(BCl_3)_{0.25}]$ and BCl_3, respectively, determined by Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES) analyses.

2. Results and Discussion

2.1. High Resolution Thermogravimetric and Modulated Differential Scanning Calorimetry Thermal Studies

The thermal stability of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes with x ranging from 0 to 0.095 is studied by high resolution thermogravimetric analyses (HR-TGA) (**Figure 1**a). Three main thermal elimination events are revealed. Event I, which starts at ≈130 °C, corresponds to the elimination of BCl_3 component. This is the result of thermal decomposition of BCl_4^- with the consequent release of hydrogen chloride.[11] Event II, detected at ≈200 °C, is attributed to the main thermal degradation of the pyrrolidinium IL. It should be pointed out that after doping the IL with δ-$MgCl_2$, event II clearly splits into two different sub-events: IIa and IIb. Finally, event III, which is absent in the sample with $x = 0$, is revealed from 260 to 280 °C. Event III is assigned to the degradation of residuals of organo-Mg species formed during events IIa and IIb. The dependence of the decomposition temperatures (T_{dec}) of each event on x is shown in Figure 1b. Results allow to reveal the presence of three different magnesium concentration regions (A, B, and C, see Table 1). This indicates that on x different magnesium and boron coordination species are formed in IL anionic nanodomains, which are likely responsible for the thermal features of the electrolytes. In A, on x, I shift toward higher temperatures. This is the result of the reaction between BCl_4^- and δ-$MgCl_2$, which yields dimer building blocks such as $[BCl_4MgCl_2]^-$. These latter are responsible for the formation of a catenated coordination network in the anionic domains of the electrolyte, which stabilizes the materials.[6a,12] In regions B and C, further doping with δ-$MgCl_2$ decreases the thermal stability of the electrolytes. Actually, a higher concentration of Mg^{2+} ions catalyzes the decomposition reaction of the IL. A similar behavior is observed for T_{IIa} and T_{IIb}, which, after an initial increase, show a slight decrease followed by a plateau.

IIa and IIb events are detected in samples with $x \geq 0.032$. In this compositional region, a high density of $[BCl_4MgCl_2]^-$ dimers is formed, which exhibit a chemical stability higher than that of $[BCl_4]^-$ units owing to the coordination of Mg^{2+} ions.

The phase and thermal transitions of the IL are investigated by modulated differential scanning calorimetry (MDSC) studies. MDSC profiles of samples, plotted in **Figure 2**a, exhibit a glass transition (T_g) in the temperature range from 58 to −52 °C. This is attributed to a structural order-disorder event occurring along pillars of ordered Pyr_{14}^+ cations neutralized by pillars of ordered chloroborate aggregates of anions.[12] At ≈−18 °C, an endothermic solid-state primary transition (T_s) appears, which is associated with the melting of very small nanodomains of

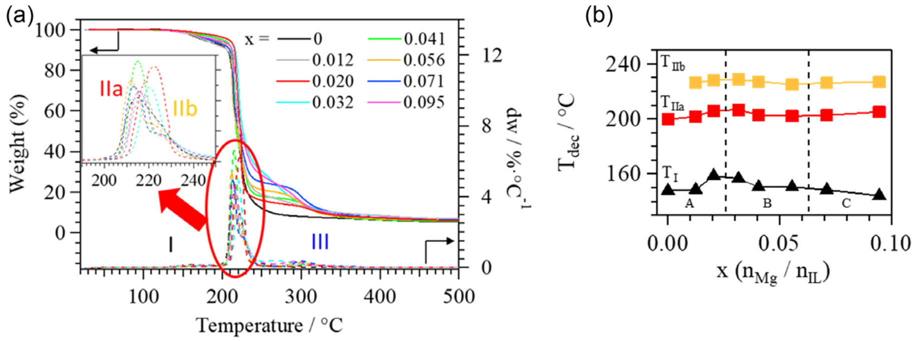

Figure 1. HR-TGA profiles of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes with x ranging from 0 to 0.095 a). A magnification of the region related to T_{IIa} and T_{IIb} thermal decompositions is shown in the inset of a). Thermal elimination temperatures T_I, T_{IIa}, and T_{IIb} are shown on x b). The graph is divided into A, B, and C concentration regions, in agreement with Table 1.

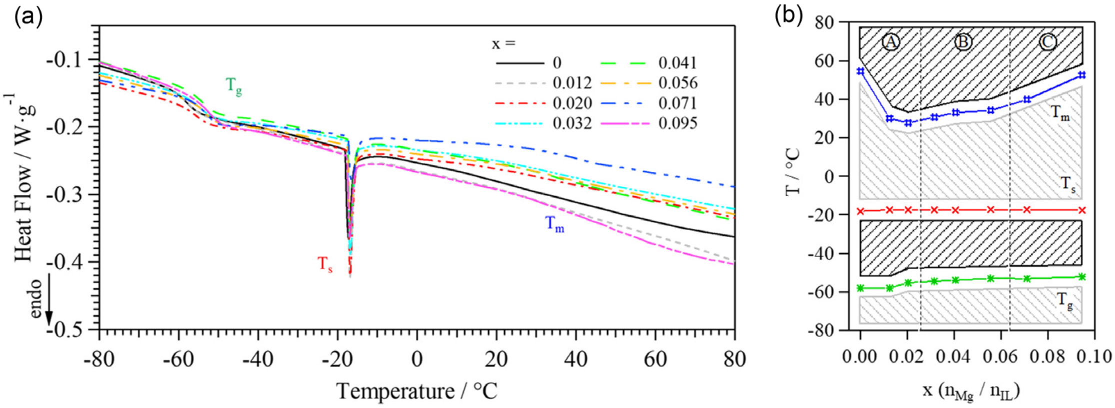

Figure 2. MDSC profiles for $[Pyr_{14}Cl/(BCl_3)_{0.5}]/(\delta\text{-}MgCl_2)_x$ samples a). Assignment of thermal transitions is reported in the figure: glass transition (T_g), solid-state primary transition (T_s), and melting transition (T_m). Dependence on x of the thermal transitions is shown in the phase diagram of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes b). The graph is divided into A, B, and C concentration regions, in agreement with Table 1.

aggregates of ordered cations. Finally, between 28 and 55 °C, a very broad melting transition (T_m) is revealed as an endothermic change in the slope of MDSC profiles, which corresponds to the formation of a viscous fluid. This, for samples at medium magnesium concentrations ($0.012 \leq x \leq 0.056$), occurs at room temperature.[6b,13]

The phase diagram on x shows that δ-$MgCl_2$ concentration (see Figure 2b) influences the measured thermal transitions. At $x > 0$, a slight increase of T_g is revealed due to the formation in anionic nanodomains of a more complex coordination network based on aggregates of catenated $[BCl_4MgCl_2]^-$ dimers.[12a] As expected, T_s, which is an endothermal transition mainly associated with nano-aggregates of cations, is not affected by the δ-$MgCl_2$ doping. The profile on x of the very weak and smooth T_m event shows that doping of the IL by δ-$MgCl_2$ leads to the formation of a highly dispersed distribution of anionic nanodomains. In B and C ($x \geq 0.032$), the distribution of anionic domains composed of different complex networks of aggregates of catenated $[BCl_4MgCl_2]^-$ dimers increases the viscosity and concurrently raises T_m as their mesoscale size grows.

2.2. Density Functional Theory Calculations

Density Functional Theory (DFT) computational studies are performed using a GGA basis set and the BLYP functionals in the DMol3 program implemented into the Materials Studio 2016 package. This is done in order to determine the optimized geometry of the components composing the IL electrolytes. Results obtained are shown in **Figure 3**. It should be noticed that: 1) the butyl group is out of plane with respect to the average plane of Pyr_{14}^+ cation ring; and 2) chloride and boron tetrachloride anions preferentially interact with

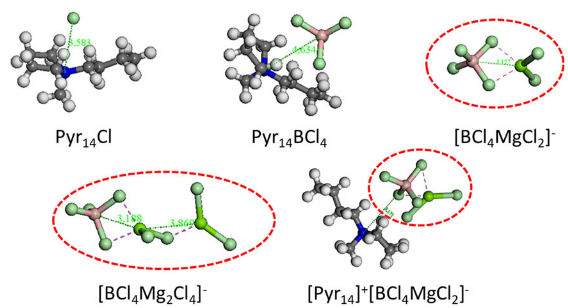

Figure 3. Coordination geometries of the different complexes present in the IL-based electrolytes. Color legend: C (black); H (white); N (blue); B (pink); Cl (light green); and Mg (green). $BCl_4^-\cdots MgCl_2$ coordination is highlighted in red.

the sterically hindered nitrogen atom of Pyr_{14}^{+} in trans to the methyl group. DFT calculations confirm that magnesium chloride units are coordinated by the anion of the IL, as already reported in the literature.[6a,12b,13] This suggests that, in proposed electrolytes, the Mg^{2+} migration occurs as a result of the fast exchange of anion species between Lewis acid sites of anionic aggregates. In addition, DFT results reveal that more than one $MgCl_2$ unit can be coordinated by each BCl_4^{-} ligand, thus yielding the typical catenated chain-like polymeric structure of the δ magnesium chloride.[14] It is important to note that the DFT-optimized geometries presented here are not intended as definitive structural assignments, but rather as chemically reasonable models that support the interpretation of experimental spectroscopic data and provide insight into the coordination trends emerging from δ-$MgCl_2$ doping.

2.3. Vibrational Spectroscopy Studies

Crucial insights on the structure and interactions in the proposed electrolytes are obtained by attenuated total reflection Fourier transform infrared (ATR-FTIR) spectroscopy studies. BCl_4^{-} anion is an interesting probe to effectively shed light by vibrational spectroscopy into: 1) the interactions and the structure of the anionic domains; and 2) the reorganization of other surrounding IL components. BCl_4^{-} shows a tetrahedral coordination geometry with a central boron atom bonded by four chlorine ligands. This highly symmetric geometry makes it an ideal probe to study the interactions and structural reorganizations responsible for the structure of the anionic and cationic domains. Docked MID and FIR ATR- FTIR spectra of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ are shown in **Figure 4**. These spectra are baseline corrected and normalized with respect to the peak centered at ≈2963 cm^{-1}, which is attributed to the antisymmetric stretching mode $\nu_a(CH_3, CH_2)$ of the pyrrolidinium butyl group. The intensity of $\nu_a(CH_3, CH_2)$ vibration is unaffected by the composition of IL electrolytes and thus it is a good internal standard for semiquantitative analyses in ATR-FTIR spectra.

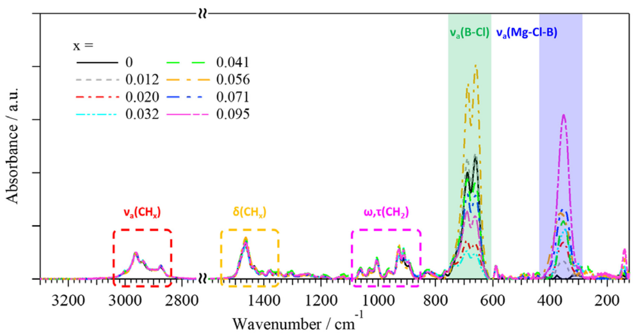

Figure 4. Docked ATR-FTIR spectra in the mid and far infrared regions of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes. The assignment of the main vibrational modes is reported in the figure. Infrared regions diagnostic of the B–Cl and Mg–Cl–B antisymmetric stretching are highlighted with a green and blue background, respectively.

In the region 3010–2800 cm^{-1}, the typical vibrational modes attributed to the ν(C–H) stretching vibrations of Pyr_{14}^{+} cation are detected, i.e., mainly the antisymmetric stretching of butyl and methyl groups $\nu_a(CH_3, CH_2)$ at 3001, 2963, 2941, 2876 cm^{-1}.[15] In the region 1500–750 cm^{-1}, peaks attributed to the bending modes of the Pyr_{14}^{+} ring $\delta(CH_2)$ and of the butyl and methyl groups $\delta(CH_2, CH_3)$ are observed at 1478 and 1380, and 1463 cm^{-1}, respectively.[15a,c] The scissoring out of plane mode of CH_3 is detected at 1434 cm^{-1}.[16] The observed wagging vibrations are: a) $\omega(CH_2)$(N) at 1405 cm^{-1}; b) $\omega(CH_2)$ at 1259, 1189, 1005, and 929 cm^{-1}; c) $\omega(CH_2)$ of the ring at 1246, 1229, and 1034 cm^{-1}; and d) $\omega(CH_2)$ of the butyl groups at 1104 and 743 cm^{-1}.[15,17] The $\tau(CH_2)$ twisting modes: 1) at 1350 and 1313 cm^{-1} are attributed to the butyl groups; 2) at 1307 cm^{-1} to the Pyr_{14}^{+} ring; and 3) at 1138 cm^{-1} to the other $\tau(CH_2)$ vibrations.[15,18] At 1122, 1062, and 911, and at 964 cm^{-1}, the typical C–C and C–N stretching vibrations are observed, respectively.[15a,b,17a] Finally, the symmetric ring breathing and rocking $\rho(CH_2)$ vibrations are peaking at 893, 829 and 765 cm^{-1}, respectively.[15a,b] In the region below 750 cm^{-1}, the typical vibrational modes of boron chloride and magnesium chloride are revealed. Peaks centered at 725, 687, 588, and 566 cm^{-1} are attributed to the ν_a(B–Cl) antisymmetric stretching vibrations.[17b,18,19] Those at 702, 634, 539, 456, 415, 376, 302, and 154 cm^{-1} are assigned to the deformation modes of Pyr_{14}^{+} cation.[15a,c,16] The stretching and bending combination modes of the B–Cl groups are detected at 659 and 496 cm^{-1}.[17b,18] In agreement with DFT calculations, the Mg–Cl antisymmetric stretching vibration, ν_a(Mg–Cl), of complexes is observed at 473 and 352 cm^{-1}. Finally, the intensities centered at 404, 315, and 274 cm^{-1} correspond to the B–Cl symmetric stretching ν_s(B–Cl), the $\nu_a + \nu_s$(B–Cl) combination mode, and the antisymmetric bending δ_a(B–Cl), respectively.[18,20] A complete correlative assignment of vibrational modes of spectra shown in Figure 4 is summarized in Table S1, Supporting Information. It is to be noted that calculated and experimental frequencies are in good agreement, confirming that the aggregates of cation and anion domains shown above as determined by DFT calculations correctly describe the real molecular and mesoscale structural features of proposed electrolytes.

In general, the spectra reported in Figure 4 reveal that the intensity and the position of vibrational modes attributed to the Pyr_{14}^{+} cation are not affected by δ-$MgCl_2$ doping of electrolytes. This, in accordance with MDSC results, confirms that pyrrolidinium cation aggregates are not influenced by x. As expected, on x, a significant modulation of the peaks assigned to the boron– and magnesium–chloride vibrational modes is observed (see Figure S2, Supporting Information). A semiquantitative analysis of the density of the B–Cl and Mg—Cl—B bonds in anion domains of bulk electrolytes is carried out, decomposing the region from 200 to 785 cm^{-1} of the ATR-FTIR spectra by Gaussian functions. This allowed us to investigate the dependence on x of both the intensity of the ν_a(B–Cl) vibrational mode peaking at 687 cm^{-1} and of the ν(Mg–Cl–B) vibration at 352 cm^{-1}. As expected, ν_a(Mg–Cl–B) peak at 352 cm^{-1} appears at $x > 0$ i.e., when $MgCl_2$ units are coordinated by BCl_4^{-} ligands. The typical ν(Mg–Cl) of pristine polymeric δ-$MgCl_2$ is, as expected, revealed at 243 cm^{-1}.[21] In agreement with the thermal analyses, three compositional regions are detected. In A at

$x \leq 0.020$, on x, a catenated coordination network is formed which increases the intensity of the ν_a(Mg–Cl–B) mode and decreases that of the ν_a(B–Cl) vibration of pristine BCl_4^-. Indeed, the formation of the $[BCl_4 \cdots MgCl_2]^-$ building blocks in anionic domains reduces the density of B—Cl bonds of pristine BCl_4^- ligands (see green line in region A of Figure S2a, Supporting Information). In B region at $0.032 \leq x \leq 0.056$, the amount of freshly added δ-$MgCl_2$ is preferentially coordinated by the already cross-linked network of $[BCl_4 \cdots MgCl_2]^-$ building blocks to $MgCl_2$ sites. This phenomenon slightly rises on x, the intensity of the ν_a(Mg–Cl–B) peak and the relative amount of $[BCl_4 \cdots MgCl_2]^-$ dimer units decreases. As a result, the typical "chain-like" structure of polymeric δ-$MgCl_2$ coordinated to BCl_4^- ligands ($[BCl_4 \cdots (MgCl_2)_n]^-$) appears (see region B of Figure S2b, Supporting Information). This situation clearly explains the trend of T_m on x as determined by MDSC measurements. Indeed, as the length of $MgCl_2$ chain in $[BCl_4 \cdots (MgCl_2)_n]^-$ complexes of the anion domains increases, the disorder in neighboring cation domains rises. This increases the mobility of the BCl_4^- ligands with a corresponding increase in the intensity of ν_a(B–Cl) mode. Taken all together, in region B, with respect to pristine $Pyr_{14}Cl/(BCl_3)_{0.25}$ IL, the cation–anion interactions are weaker and the disorder of the ionic aggregates is higher (i.e., T_m of electrolytes is lower than that of pristine $Pyr_{14}Cl/(BCl_3)_{0.25}$ sample, see MDSC studies). In region C, at $x \geq 0.071$, $Pyr_{14}Cl/(BCl_3)_{0.25}$ IL is saturated with δ-$MgCl_2$, forcing the formation of $[BCl_4 \cdots MgCl_2]^-$ interactions which results in a decrease of the intensity of the mode ν_a(B–Cl) and in an increase of the intensity of the ν_a(Mg–Cl–B) vibrational mode (see Figure S2a, Supporting Information).

2.4. Broadband Electrical Spectroscopy Studies

The electrical properties of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes are studied in depth by broadband electrical spectroscopy (BES) technique with the aim to: 1) reveal the conductivity mechanisms which give rise to the long-range charge migration processes; 2) detect the correlation occurring between the structure of the ionic aggregates and the charge transfer mechanisms; and 3) study the effects of the δ-$MgCl_2$ doping on the overall electric response of the IL-based electrolytes in terms of dielectric relaxation and polarization phenomena.[7a,22] BES measurements are carried out in the frequency range from 3×10^{-2} to 10^7 Hz, from −100 to 100 °C. The 3D spectra of imaginary component of permittivity on temperature and frequency of selected electrolytes with $x = 0$, 0.020, 0.041, and 0.071 are shown in **Figure 5** (see Figure S3, Supporting Information for results of the other samples). A qualitative analysis of these spectra allows for to detection of the presence of an electrode polarization (σ_{EP}) and up to three interdomain polarizations (σ_{IP_k}, with $1 \leq k \leq 3$) events (see Figure 5 and S3, Supporting Information). σ_{EP} is attributed to the charge accumulation at the electrode/electrolyte interface, while σ_{IP_k} correspond to the accumulation of charges at the interfaces between nanodomains in bulk electrolytes with different permittivity. The presence of σ_{IP_k} phenomena clearly confirms that the $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes in the mesoscale are heterogeneous. At high frequency, two α_j and two β_j dielectric relaxations are detected with $j = 1,2$ (see Figure 5 and S3, Supporting Information).

The electric response, in terms of both the imaginary and real components of the complex conductivity ($\sigma^*(\omega)$) and complex permittivity ($\varepsilon^*(\omega)$), was analyzed as elsewhere reported by Equation (1):[7a]

$$\varepsilon^*(\omega) = -i\left(\frac{\sigma_0}{\omega\varepsilon_0}\right)^N + \frac{\sigma_{EP}(i\omega\tau_{EP})^{\gamma_{EP}}}{i\omega\varepsilon_0[1+(i\omega\tau_{EP})^{\gamma_{EP}}]} + \sum_{k=1}^{n}\frac{\sigma_{IP,k}(i\omega\tau_{IP,k})^{\gamma_{IP,k}}}{i\omega\varepsilon_0[1+(i\omega\tau_{IP,k})^{\gamma_{IP,k}}]} + \sum_{j=1}^{m}\frac{\Delta\varepsilon_j}{[1+(i\omega\tau_j)^{a_j}]^{b_j}} + \varepsilon_\infty \tag{1}$$

$$\varepsilon^*(\omega) = \varepsilon'(\omega) - i\varepsilon''(\omega) \tag{2}$$

$$\sigma^*(\omega) = i\omega\varepsilon_0\varepsilon^*(\omega) \tag{3}$$

$$\omega = 2\pi f \tag{4}$$

The first term of Equation (1) describes the material conductivity at zero frequency. ε_∞ accounts for the permittivity of the material at infinite frequency (electronic contribution). The second term is associated with the electrode polarization event, and the third describes the interdomain polarization phenomena. σ_k and τ_k are the conductivity and the relaxation times, respectively, associated with the kth electrical event. γ_k is attributed to the broadening of the kth polarization peak. The third term accounts for the dielectric relaxation phenomena.[7a] $\Delta\varepsilon$, τ_j, α_j, and β_j parameters correspond to the dielectric strength, relaxation time, symmetric and antisymmetric shape parameter of each jth relaxation event, respectively.

2.4.1. Polarization Phenomena

σ_{EP} and σ_{IP_k} (with $1 \leq k \leq 3$) parameters, determined as above described by fitting simultaneously components of $\sigma^*(\omega)$ and $\varepsilon^*(\omega)$ complex spectra by Equation (1), are plotted on T^{-1} in **Figure 6**. The overall conductivity (σ_T) is the superposition of σ_{EP} and σ_{IP_k} conductivity pathways with $\sigma_{EP} >> \sigma_{IP_k}$. Four temperature regions are detected in Figure 6 (I, II, III, and IV), which are delimited by the thermal transitions revealed by MDSC studies. The presence of σ_{IP_k} demonstrates that the proposed materials in mesoscale present the heterogeneities discussed above on the basis of DFT, FTIR, and MDSC studies, i.e., IL electrolytes likely consist of domains of pyrrolidinium cation aggregates neutralized by anion nanodomains based on catenated linear magnesium chloroborate complexes. At $T > T_g$, σ_s, with $s = $ EP, IP,k show a Vogel–Tammann–Fulcher (VTF)-like behavior, indicating that the conductivity mechanism along these percolation pathways is assisted by the relaxations of the IL host matrix. At $T < T_g$ (Region I), only the electrode polarization event with a very low conductivity ($<10^{-13}$ S cm^{-1}) is revealed, which indicates that in I the mobility of ionic species in anionic domains is inhibited. This Arrhenius-like behavior demonstrates that in I, the long-range charge migration occurs through hopping of ions between neighboring coordination sites in anion domains. These charge migration phenomena are decoupled from dielectric relaxation events.

The contribution of each conductivity pathway to the overall conductivity, determined by Equation (5):

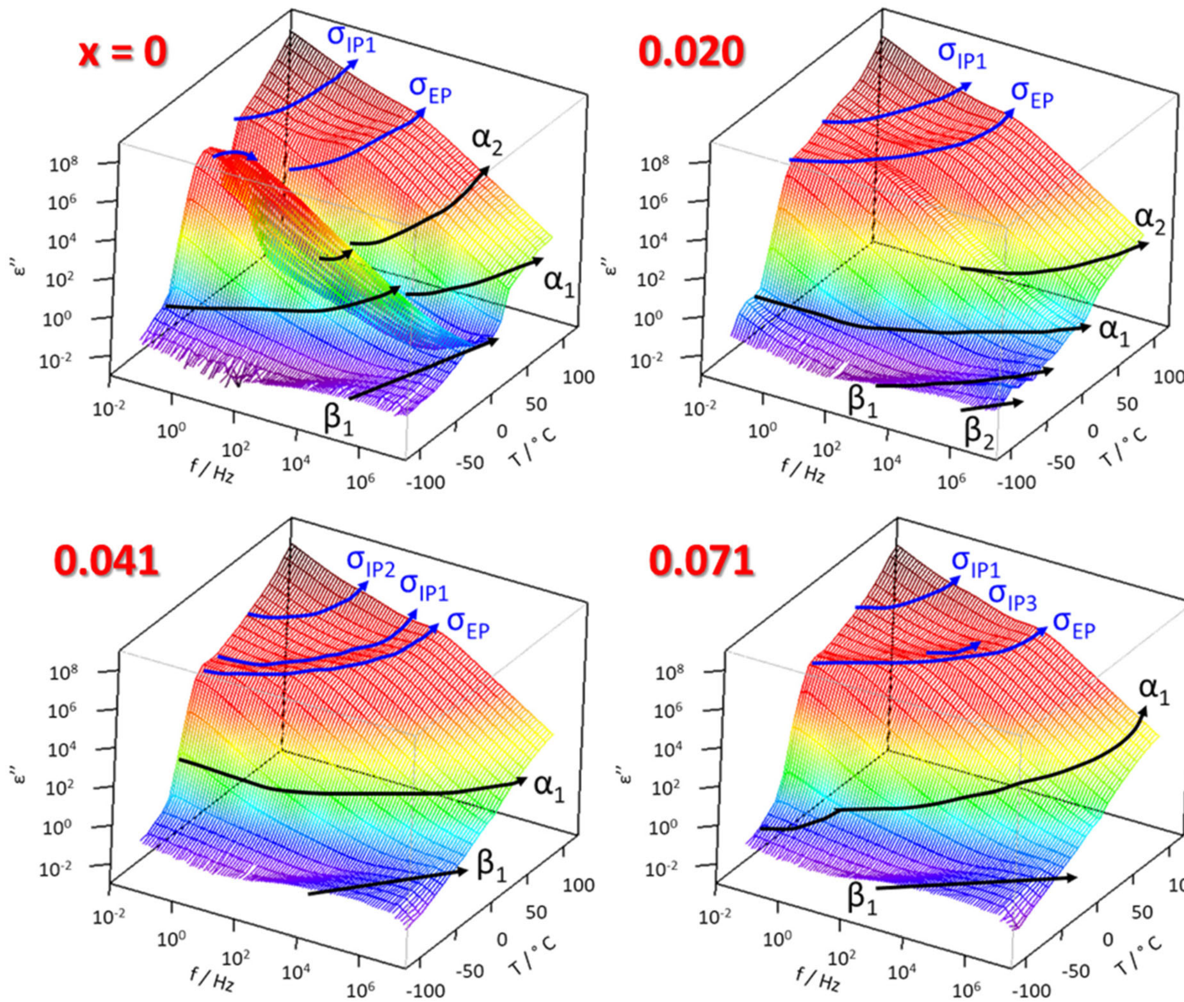

Figure 5. 3D spectra of the imaginary component of the permittivity on frequency and temperature for selected $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes, with $x = 0$, 0.020, 0.041, and 0.071. The detected polarization phenomena (σ_s, with s = EP, IP,k with $k = 1$, 2, and 3) and dielectric relaxation events (f_j, with $j = \alpha_1$, α_2, β_1, β_2) are reported in the 3D spectra.

$$\varphi_s = \frac{\sigma_s}{\sigma_{EP} + \sum_{k=1}^{3} \sigma_{IP,k}} \tag{5}$$

with s = EP, IP,k (see Figure S4, Supporting Information) shows that σ_{EP} is dominating the overall conductivity of electrolytes in almost the whole studied temperature range. Only $\sigma_{IP,1}$ exhibits values with an order of magnitude like that of σ_{EP} in II. The overall conductivity (σ_T) is determined by Equation (6) and is the result of the superposition of all the anionic percolation pathways, σ_s, with s = EP, IP,k, and k = 1, 2, and 3, present in materials:[7a]

$$\sigma_T = \sigma_{EP} + \sum_{k=1}^{3} \sigma_{IP,k} = \sigma_{EP} + \sigma_{IP,1} + \sigma_{IP,2} + \sigma_{IP,3} \tag{6}$$

Figure S5, Supporting Information shows the dependence of σ_T on T^{-1} of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes in regions A, B and C. Pure IL and samples with a low magnesium concentration (region A) show a drop in conductivity at $T_s \leq T \leq T_m$, which is gradually inhibited as the salt concentration increases. This event is assigned to the structural reorganization that occurs during T_s.[12b,23] Indeed, as previously hypothesized (see MDSC results), at T_s occurs the melting of small nanoaggregates of Pyr_{14}^+ cations, which partially compromises the connectivity of catenated anionic domains, inhibiting the exchange of anion ligands. This happens owing to local electrostatic anion-Pyr_{14}^+cation interactions. This effect on x is gradually attenuated due to the increase of the density of $[BCl_4MgCl_2]^-$ coordination anion species, which starts consolidating the connectivity of catenated complexes in anionic domains (see MDSC studies). This, above T_s transition, results in the formation of larger and partially connected conductivity pathways, which yield the lowest values of σ_T, i.e., up to 6.08×10^{-5} S cm^{-1} at 40 °C. In B, no drop of σ_T on x is detected. This witnesses that in this region, doping of electrolytes with δ-$MgCl_2$ consolidates significantly the long-range connectivity of catenated complexes in anionic domains, giving rise to efficient anionic percolation pathways for conductivity. Thus, at 40 °C,

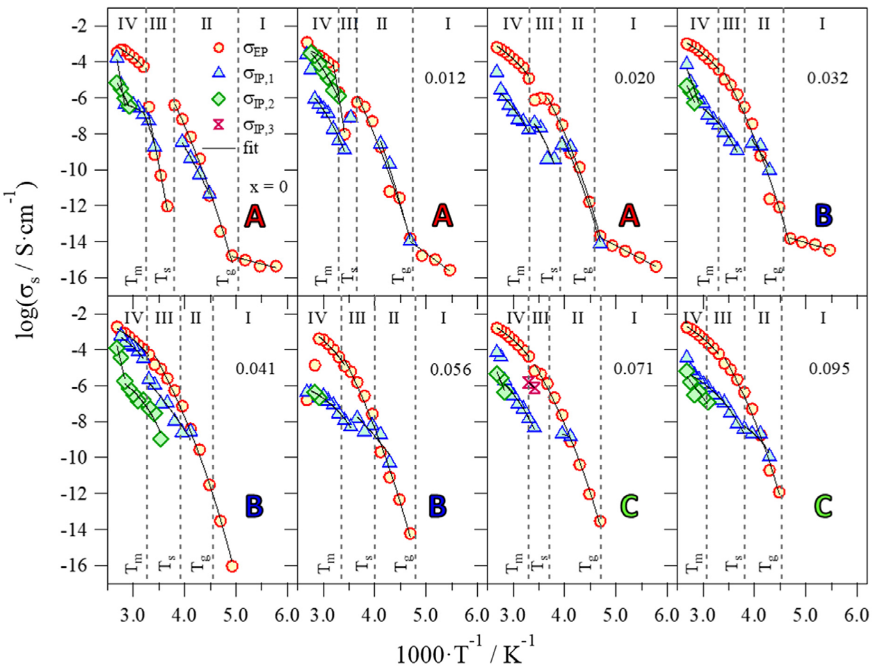

Figure 6. Log σ_s versus T^{-1} (s = EP, IP,k with k = 1, 2 and 3) profiles of polarization events of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes obtained by fitting 3D BES spectra with Equation (1). Temperature regions I, II, III, and IV are shown corresponding to the transition temperatures measured by MDSC.

$[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_{0.041}$ electrolyte exhibits the highest value of ionic conductivity of 1.43×10^{-4} S cm^{-1}. In C, on x, the high magnesium concentration gradually increases the density of crosslinks in anion aggregates, which limit σ_T values and raise the viscosity of electrolytes. In this region, at 40 °C, $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_{0.095}$ electrolyte shows an overall conductivity of 1.24×10^{-4} S cm^{-1}. These ionic conductivity values are comparable with the state-of-the-art Mg^{2+}-conducting electrolytes, especially considering the higher thermal stability compared with organic liquid-based systems (see **Table 2**).

2.4.2. Dielectric Relaxation Events

To disclose the role played by the dielectric relaxation modes to the electric response of the electrolytes, the dependence on T^{-1} of the frequency (f_j) of the dielectric modes is studied (**Figure 7**a). Four different dielectric relaxation events (f_j, with $1 \leq j \leq 4$) are revealed: 1) two α-modes (α_1 and α_2); and 2) two β-modes (β_1 and β_2). α_1 and α_2 are associated with the diffusion along the stacking axis of distortional motions around the short and the long axis of spheroidal cation molecules (β_1 and β_2), respectively. α_2 is correlated to diffusion of orientational states of interacting butyl chains of neighboring cations along the direction parallel to the stacking axis (see Figure 7b).[6b,12a] As expected, β_2 is revealed at lower frequencies with respect to β_1, due to the steric hindrance of the ordered butyl chains. On T^{-1}, α_1 shows an Arrhenius-like behavior at $T < T_g$, while a VTF one at $T > T_g$. This confirms that, in correspondence with the T_g thermal secondary transition, significant structural changes take place both in anionic and cationic domains.

At $0 < x < 0.032$, α_2 gradually shifts to high frequencies on doping. This indicates that an increasingly faster long-range diffusion of orientational states of butyl chains between neighboring interacting cations along the stacking axis of cationic aggregates takes place, i.e., an efficient long-range inter-chain coupling of the motion of butyl side chains governs α_2 mode.

At $x \geq 0.032$, no α_2 and β_2 modes are detected. α_2 and β_2 are shifted out of the experimentally measured range at higher frequencies, i.e., a high concentration of $\delta\text{-}MgCl_2$ in anionic domains influences significantly both the long-range coupling and the local orientational motions of butyl side chains of cationic aggregates. This, in accordance with other studies on ILs,[6,12,13] unequivocally confirms that the structure of anion domains of proposed ILs is consistent with catenated 3D coordination networks of metal halide units. As expected, in the explored temperature range, β relaxations show Arrhenius-like behaviors, thus confirming the correctness of spectral assignments.[7a]

2.4.3. Activation Energies of Conductivity Pathways and Dielectric Relaxations

Fitting of curves of Figure 6 and 7a by VTF,[24] Vogel–Tammann–Fulcher–Hesse,[7a] or Arrhenius-like (A) equations[7a] allowed to determine the values of pseudoactivation energy (E_a) of conductivity pathways and dielectric relaxation events. Results shown in Figure S6, Supporting Information, evidence that the overall conductivity (σ_T) of ILs corresponds to the superposition of the conductivity pathways σ_s, with s = EP, IP,k, and k = 1, 2, and 3. It should be noticed that the dominant contribution to σ_T is given by σ_{EP}. This is predominantly correlated to the α_1 dielectric relaxation at all temperatures and sample compositions. However, a correlation is also observed between: 1) σ_{EP} and α_2 for $x \leq 0.020$; and 2) $\sigma_{IP,1}$ and α_1 for samples with $x \geq 0.032$.

Table 2. Ionic conductivity and thermal stability of some of the state-of-the-art Mg^{2+}-conducting electrolytes.

Type	Salt	Solvent	σ at r.t. [S cm^{-1}][a)]	Thermal stability [°C]	References
Liquid	$Mg(CB_{11}H_{12})_2$	TEGDME	1.8×10^{-3}	190	[29]
	$Mg[B(HFP)_4]_2$	DME	6.8×10^{-3}	150	[30]
	$Mg(AlCl_2BuEt)_2$	THF	$\approx10^{-3}$	–	[31]
	HMDSMgCl-$MgCl_2$	THF	2×10^{-4}	–	[32]
	PhMgCl-$AlCl_3$	THF	4×10^{-3}	150	[33]
	$MgCl_2$-$AlCl_3$	DME	2×10^{-3}	–	[34]
Ionic liquid	$Mg(TFSI)_2$	MMPITFSI	1.75×10^{-3}	–	[35]
	$Mg(TFSI)_2$	PYR_{14}TFSI	2.2×10^{-3}	300	[36]
	$Mg(TFSI)_2$	PYR_{1201}TFSI	3.2×10^{-3}	300	[36]
	δ-$MgCl_2$-$AlCl_3$	EMImCl	5.5×10^{-3}	300	[6a]
	δ-MgI_2-AlI_3	EMImI	3.0×10^{-4}	–	[12b]
	δ-$MgCl_2$-$TiCl_4$	EMImCl	2.5×10^{-4}	–	[13]
	δ-$MgCl_2$-$AlCl_3$	PYR_{14}Cl	2.1×10^{-3}	–	[12a]
	δ-$MgCl_2$-Me_2SnCl_2	PYR_{14}Cl	1.2×10^{-3}	–	[6b]
	δ-$MgCl_2$-BCl_3	PYR_{14}Cl	1.4×10^{-4}	200	This work
	$MgCl_2$	$BMIMPF_6$	4.6×10^{-7}	–	[37]
	$Mg(BH_4)_2$	PYR_{14}TFSI	–	–	[38]
	$Mg(BH_4)_2$	PYR_{1201}TFSI	–	–	[38]
	$Mg(BH_4)_2$	PYR_{120201}TFSI	–	–	[38]
	$Mg(BH_4)_2$	N_{01}TFSI	–	–	[38]
	$Mg(BH_4)_2$	N_{02}TFSI	5.7×10^{-4}	–	[38]
	$Mg(BH_4)_2$	N_{07}TFSI	4.3×10^{-4}	–	[38]

a)Temperature is reported in brackets when the conductivity is not referred to room temperature.

In addition, β_1, i.e., the local fluctuation of Pyr_{14}^+ cations around the short axis, appears to contribute to modulation of the long-range charge migration process in the temperature range $T_g \leq T \leq T_s$. Taken all together, it is determined that conductivity pathways of magnesium chloroborate complexes are clearly coupled with dielectric relaxation events of cation aggregates. This coupling phenomenon depends on temperature and magnesium concentration, and preferentially is correlated to the α_1 mode. In conclusion, in proposed IL electrolytes, the overall conductivity is the result of long-range migrations of anions between catenated anionic domains mediated by segmental motions of cation aggregates.

2.4.4. Diffusion Coefficient and Average Migration Distance

Additional information on the long-range charge transfer mechanisms modulating the conductivity of proposed electrolytes are determined by studying carefully the temperature dependence of both the diffusion coefficients [$D(\sigma_s, T)$ with $s = $ EP, IP,k and k = 1, 2 and 3] and the average charge migration distance ($\langle r(\sigma_s, T)\rangle$) of electrolytes. $D(\sigma_s, T)$ is calculated using the Nernst–Einstein Equation (7):[7a]

$$D(\sigma_s,\ T) = \frac{RT\sigma_s}{Z_n^2 C_n F^2} \tag{7}$$

where R is the gas constant, T is the temperature at which D is determined, σ_s is the conductivity of the s = EP, IP,k and k = 1, 2 and 3 polarization, Z_n and C_n correspond to the charge and concentration of the nth species migrating along the σ_s conductivity pathway, and F is the Faraday constant.

To detect how dielectric relaxations of host materials are modulating the long-range charge migration processes in electrolytes, $\log(D(\sigma_s, T))$ is correlated to $\log(f_j)$ of dielectric relaxations. Results shown in Figure S7a, Supporting Information confirm that both σ_{EP} and $\sigma_{IP,1}$ pathways are correlated to α_1 dielectric relaxation, while only σ_{EP} is correlated to α_2. This witnesses that the segmental motions along the aggregation axis of both stacking rings and interacting butyl side groups of cations play a crucial role in modulating the overall conductivity mechanism of materials. β_1 mode is contributing only slightly to the stimulation of σ_{EP} conductivity pathway.

The average charge migration distance ($\langle r(\sigma_s, T)\rangle$) along the σ_s conductivity pathways is determined by using the Einstein–Smoluchowski Equation (8):[7a]

$$\langle r(\sigma_s,\ T)\rangle = \sqrt{6D(\sigma_s,\ T)\,\tau_s} = \sqrt{\frac{6RT\sigma_s\,\tau_s}{Z_n^2 C_n F^2}} \tag{8}$$

where τ_s is the relaxation time, σ_s the conductivity pathway s and $D(\sigma_s, T)$ is given by Equation (7). Results reveal that mainly the average migration distance of sth pathway increases with temperature (see Figure S7b, Supporting Information). In detail, at $T \leq -60$ °C (region I), no migration of species is revealed. In the range $-60 \leq T \leq -15$ °C (region II), on T $\langle r(\sigma_{EP}, T)\rangle$ increases from 0.01 nm up to 100 nm, i.e., by almost four orders of magnitude. In these temperature regions, $\langle r(\sigma_{EP}, T)\rangle$is not affected by x. In III ($-15 \leq T \leq 30$ °C), the conductivity of sth pathways exhibits $\langle r(\sigma_s, T)\rangle$ values of the same order of magnitude. On x, only a slight increase of $\langle r(\sigma_s, T)\rangle$ is registered, likely associated with the structural distortions generated in the catenated anionic domains by the coordination of $MgCl_2$ units. At $T \geq 30$ °C (region IV), $\langle r(\sigma_s, T)\rangle$ shows the highest values. This demonstrates that in these conditions, delocalization volumes[25] with large sizes are generated in 3D catenated anion domains owing to fast anion exchange processes between Lewis acid coordination sites.

2.5. Conductivity Mechanism

BES studies provide crucial insights to shed light on the phenomena influencing the long-range charge migration events characterizing the overall conductivity of proposed electrolytes, both on temperature and x. Indeed, BES results exhibit multiple dielectric relaxation processes that reflect the complex ion dynamics and mesoscale structuring of the system (see Figure 6). The low-frequency response is dominated by a strong electrode polarization (σ_{EP}) effect, which arises from the accumulation of charge carriers at the electrode/electrolyte interface. This phenomenon is commonly observed in IL and polymeric systems where ion mobility is limited by confinement or interfacial

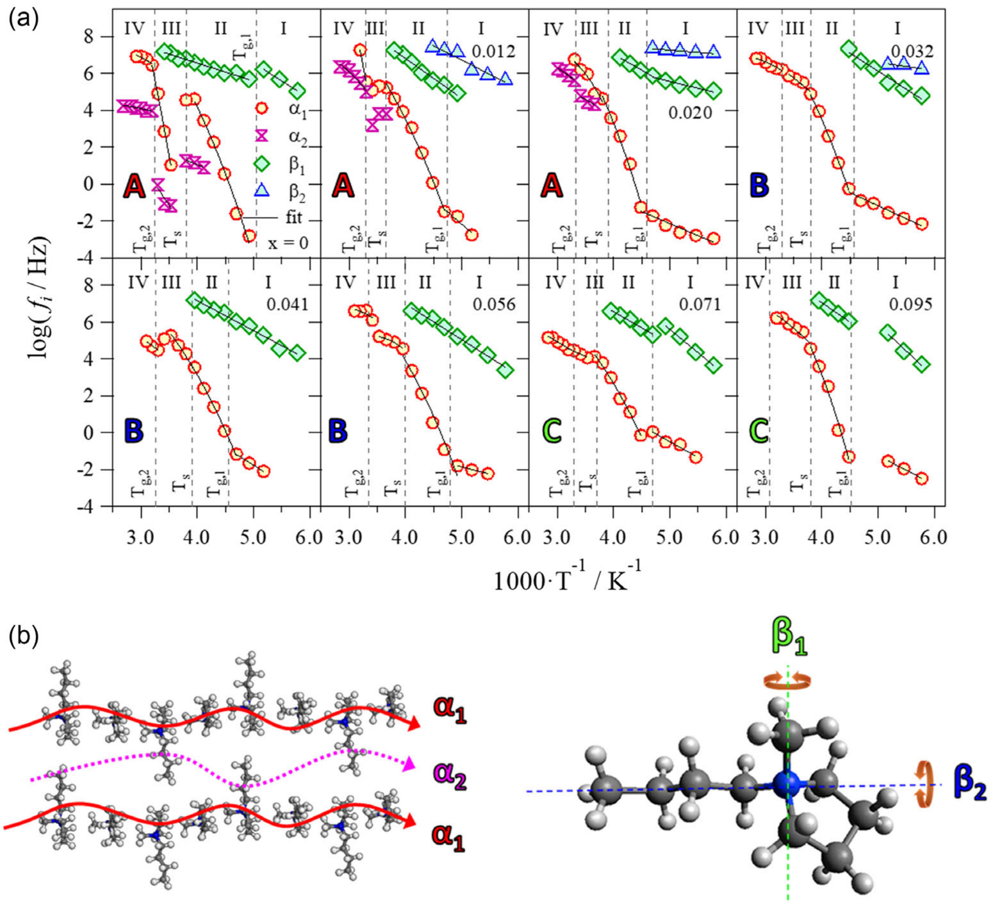

Figure 7. Log f_j versus T^{-1} profiles of f_j dielectric relaxation events of IL electrolytes obtained by fitting 3D BES spectra with Equation (1) a). Four temperature regions (I, II, III, and IV) are revealed, which are delimited by the temperature of thermal transitions revealed by MDSC studies. Scheme showing the type of dielectric relaxation events (i.e., α_1, α_2, β_1, β_2) detected in proposed materials b).

interactions.[6a,7a,25] At intermediate frequencies, distinct interdomain polarization ($\sigma_{IP,k}$) processes are observed, which are associated with charge accumulation at the interfaces between nanoscale ionic domains with different permittivity. These processes reflect the permittivity contrast and dynamic heterogeneity between regions rich in $[Pyr_{14}]^+$ cations and those dominated by $[BCl_4MgCl_2]^-$ complexes. According to the formalism developed by Di Noto and coworkers,[7a] such interfacial relaxation processes can be rationalized using a "multiphase dielectric model" in which the electrolyte is treated as a heterogeneous system composed of dynamically interconnected domains with different dielectric constants and conductivities. The IP,k phenomena are thus fingerprints of the mesoscale morphology and the dynamic percolation of charge carriers through the evolving coordination network. The frequency dispersion and temperature dependence of these features suggest that the overall conductivity is governed not only by the mobility of free ions and charged complexes, but mainly by the segmental motion of dynamically cross-linked anion-cation networks. This dual mechanism transport along percolating ionic pathways (long range) and through localized hopping between coordination sites (short range) is consistent with previous studies on functionalized polymer and IL-based systems.[6a,7a,25]

In more detail, results demonstrate that at low temperatures (region I), the only possible conductivity mechanism is the "hopping" of anion species such as Cl^-, $[BCl_4MgCl_2]^-$, and others between different Lewis acid coordination sites. This is supported by the very small average migration distance values observed in this region. At the medium-low temperatures (region II), the IP,1 pathway significantly contributes to the overall conductivity. This outcome indicates that in the studied electrolytes, the presence of domains with different permittivity, i.e., cation and anion aggregate nanodomains as confirmed by DFT, FTIR, and MDSC studies, plays a crucial role in long-range migration events at the interfaces between cation and anion aggregate nanodomains ($\sigma_{IP,1}$). The conductivity of σ_{EP} is modulated by α_1 and α_2 modes, while that of $\sigma_{IP,1}$ is assisted only by α_1 dielectric relaxation. α_1, which corresponds to an efficient diffusion of distortional motions along the axis of cation aggregates, is very effective in stimulating the exchange of the anion species between different coordination sites within anionic domains. Indeed, in II, on T, a steep increase in the migration distance up to 90 nm is observed for the conductivity pathways.

In catenated anionic domains, the fast exchange of anion species between coordination sites, with respect to the slower time scale of overall conductivity, yields 3D domains within which the ions can be considered delocalized, i.e., as elsewhere defined delocalization Bodies (DBs)[25] of anionic domains. σ_{EP}, which is significantly coupled with α_1 mode, occurs owing to the exchange of anion species between different neighboring DBs.

At medium-high temperatures (region III), a drop in the conductivity is observed for the samples at low magnesium salt concentration ($x < 0.032$), which is attributed to the T_s thermal transition. Likely, in these conditions, the order-disorder transition of cation domains partially reduces the diffusion of distortional motions along the aggregation axis of cation aggregates (α_1 mode), thus inhibiting the long-range exchange of anions between DBs. At $x \geq 0.032$, the further excess of δ-$MgCl_2$ is better coordinated by Lewis base sites in catenated anionic domains, increasing the size of DBs, which thus improves the coupling of σ_{EP} with α_1 mode and raises the average migration distance up to ≈90 nm. A further increase of the size and of the mobility within the DBs is registered in region IV, where the IL electrolytes are viscous liquids exhibiting: 1) an average migration distance up to 500 nm; and 2) conductivity pathways significantly coupled with α_1 and α_2 dielectric relaxations, as demonstrated by their similar E_a values of ≈6 kJ mol^{-1}.

It must be clarified that within each DB, active ions undergo rapid local exchange between coordination sites formed by $[BCl_4MgCl_2]^-$ and related species. This intradomain exchange occurs on the time scale captured by BES measurements and supports efficient localized ion motion. Critically, a fast exchange within DBs is a necessary condition for enabling long-range charge migration. However, the rate-determining step for such long-range transport lies in the exchange of active ions between neighboring DBs, a process governed by the mesoscale organization of the nanostructured electrolyte. This behavior is consistent with the relatively large size of the DBs, as inferred from the average charge migration distances reported in Section 2.4.4. The exchange of active ions between adjacent DBs occurs only when the dynamics of cationic and anionic nanodomains are sufficiently coupled a phenomenon that governs the material's macroscopic viscosity. Thus, viscosity in this system is not a primary transport barrier, but an emergent property of the underlying mesoscale organization and domain dynamics. BES analysis confirms that these DB-mediated percolation pathways are operative and strongly temperature-dependent. Altogether, the concept of DBs offers a mechanistic framework to rationalize the observed decoupling between structural viscosity and ionic mobility in these complex IL-based electrolytes.

2.6. Nuclear Magnetic Resonance Studies

Nuclear magnetic resonance (NMR) studies are performed in order to evaluate the ion speciation and the chemical environment of the different domains composing the IL-based electrolytes. Four samples at different magnesium concentrations ($x = 0$, 0.032, 0.056, and 0.095) are selected for NMR studies as representatives of all the proposed materials.

2.6.1. 1D ^{1}H and ^{11}B NMR

Results of 1D ^{1}H NMR studies are shown in Figure S8a, Supporting Information. Proton nuclei are present only in the Pyr_{14}^+ cation, which is characterized by protons in seven different chemical environments.[12a,26] On x, no shifting of ^{1}H peaks is observed, indicating that the shielding of protons in Pyr_{14}^+ cations is not affected by the different anionic species that are forming upon addition of δ-$MgCl_2$. This is in agreement with MDSC and FTIR studies, which clearly demonstrated that in these electrolytes the cation nanodomains are only marginally affected by the evolution of the coordination network established in the anion nanodomains.

Room temperature 1D ^{11}B NMR spectra (see Figure S8b, Supporting Information) reveal the presence of only one peak centered at ≈−12.9 ppm. This signal is attributed to the $[BCl_4]^-$ species composing the anion nanodomains.[27] Interestingly, in Mg-doped electrolytes, a downfield shift of the ^{11}B signal is observed (≈0.025 ppm), indicating that the formation of $[BCl_4MgCl_2]^-$ dimers induces a deshielding effect on boron nuclei. The absence of a second peak in 1D ^{11}B NMR spectra of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes (with $x > 0$) indicates that the chemical exchange between $[BCl_4]^-$ and $[BCl_4MgCl_2]^-$ species within DBs is very fast, and magnesium chloride units can be considered delocalized on the timescale of the NMR spectroscopy. In addition, the presence of a single peak in all the 1D ^{11}B NMR spectra confirms that free BCl_3 molecules do not exist in this IL-based electrolyte.[27] This indicates that boron trichloride is quantitatively complexed by Lewis basic ligands of catenated anion domains of the pristine $Pyr_{14}Cl$ IL, such as $[BCl_4]^-$ and $[BCl_4MgCl_2]^-$ species. On temperature (see Figure S8c, Supporting Information), ^{11}B peaks become sharper and deshielded, pointing out the high mobility of boron-based species and the decrease of density of dynamic crosslinks inside the anionic catenated domains. Thus, at high temperature, faster mobilities of molecular species (*i.e.*, faster rotational and translational dynamics in anionic domains) are expected.

2.6.2. 1D ^{25}Mg NMR

Insights into the coordination of magnesium ions are obtained through ^{25}Mg NMR studies, investigating the spectra of electrolytes on x and temperature (see **Figure 8**). The NMR analysis reveals a single ^{25}Mg peak, as expected. The chemical shift for this peak is ≈49.6 ppm with respect to an aqueous solution of 11 M $MgCl_2$. This result is very interesting when compared to the analogous IL-based electrolytes where boron is substituted by aluminum ($[Pyr_{14}Cl/(AlCl_3)_{1.5}]/(\delta\text{-}MgCl_2)_x$).[12a] For these latter, the observed ^{25}Mg chemical shift is ≈1.5 ppm. These evidences indicate that the presence of boron in the coordination network has a strong electron-withdrawing effect on the magnesium nuclei, more intense with respect to aluminum. Thus, stronger Mg···Cl···B coordination interactions are expected compared to those formed in chloroaluminate analogs. The ^{25}Mg linewidth becomes narrower on T, as depicted in Figure 8d, while it is constant on x. The narrowing phenomenon suggests an enhanced mobility of Mg-based coordination species and a decrease in the density of dynamic crosslinks within the anionic aggregates as the temperature rises, in agreement with BES studies. The constant FWHM on x reveals that the increasing doping

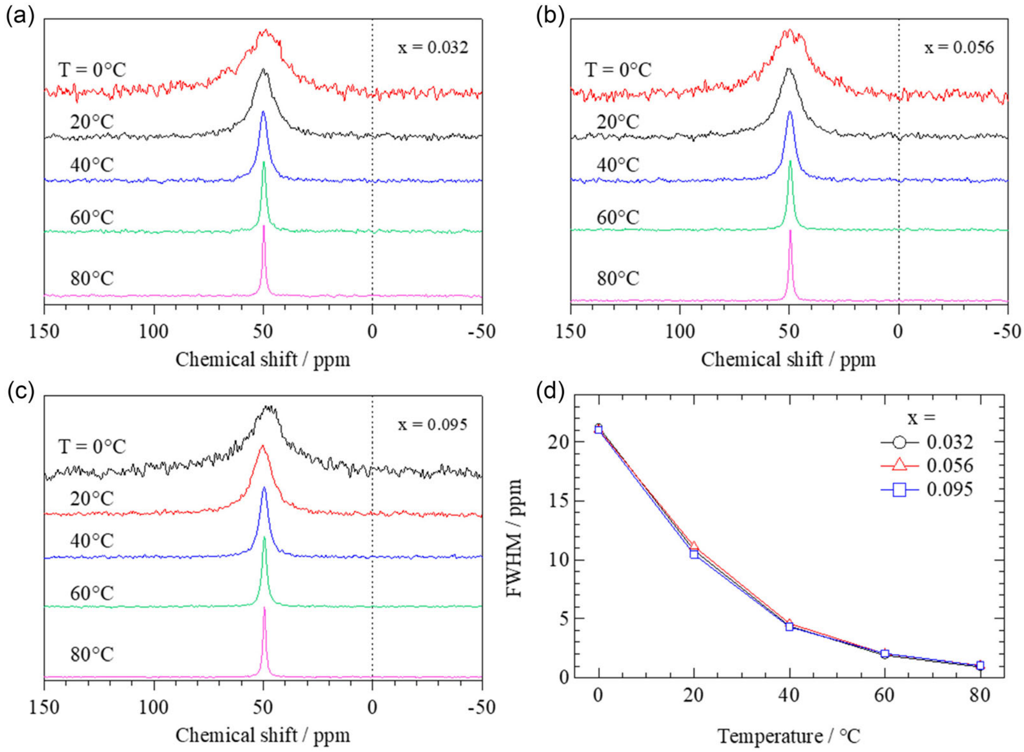

Figure 8. ^{25}Mg Hahn-echo NMR spectra of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ samples recorded at static condition as a function of temperature. Results for $x = 0.032$ a), 0.056 b), and 0.095 c). Results of peak fitting to one component (Lorentzian line shape) as a function of temperature for the three samples d).

with δ-$MgCl_2$ acts on the concentration of $[BCl_4MgCl_2]^-$ species, and not on the strength of Mg···Cl···B coordination interactions.

2.6.3. PFG ^1H and ^{11}B NMR

^1H and ^{11}B PFG-NMR studies are performed to investigate the contribution of each single ionic species to the overall charge diffusion (D) in the proposed materials. In detail, ^1H self-diffusion coefficients are determined to quantify the role of Pyr_{14}^+ ions in cationic domains and in the long-range charge migration process, while ^{11}B D values highlight the contribution of magnesium chloroborate active species. The D values for all the samples at every temperature are determined first by calculating the integrated intensities (S) from each experiment as a function of the applied gradient (g), and then using a least-squares fitting of the Stejskal–Tanner Equation (9):[28]

$$S = S_0 e^{-D(g\delta\gamma)^2\left(\Delta - \delta/3\right)} \tag{9}$$

where δ corresponds to the gradient pulse duration, Δ is the diffusion delay, and γ stands for the gyromagnetic ratio of the ^1H or ^{11}B nuclei. The dependence on T of D values for both ^1H and ^{11}B is shown in **Figure 9**. The self-diffusion coefficient for ^1H is on the order of 10^{-10} cm^2 s^{-1} at low temperatures (i.e., 0 °C) and rises up to 10^{-7} cm^2 s^{-1} at high temperatures (i.e., 100 °C) (see triangle markers in Figure 9). Higher values are obtained for ^{11}B, ranging from 10^{-9} to 10^{-7} cm^2 s^{-1} going from low to high temperature. This difference is more evident below T_m (i.e., $T < 40$ °C) and for intermediate Mg concentration region B (i.e., $x = 0.032$ and 0.056). Taken all together, all the studies performed in this work on the proposed materials indicate a nonlinear behavior of x of the physico-chemical properties of the investigated electrolytes. The ion speciation and the coordination network that is formed in IL-based matrix have a fundamental role also in the dynamics and long-range charge migration of the active species. Above T_m (i.e., $T > 40$ °C), a more disordered system mitigates these differences on x. The relative contribution of cation species (i.e., D_{1H}) to the total diffusion coefficient (i.e., the sum of Pyr_{14}^+ cation and magnesium chloroborate anion diffusions) is highlighted by calculating the proton diffusivity fraction (u_H) using Equation (10) (see Figure 9b):

$$u_H = \frac{D_{1_H}}{D_{1_H} + D_{11_B}} \tag{10}$$

A parabolic behavior on T is observed for all the samples, except for the highest concentrated sample ($x = 0.095$) at low temperature ($T \leq 20$ °C), where probably the high viscosity becomes more relevant. This trend is explained considering that at low temperature, the cation nanodomains are well organized and dynamically hindered. In the same temperature region, boron-based species are more dynamic, and chloroborate/magnesium chloroborate complexes are easily exchanged within the anionic nanodomains. Thus, at low temperature, boron-based species show a higher diffusion coefficient with respect

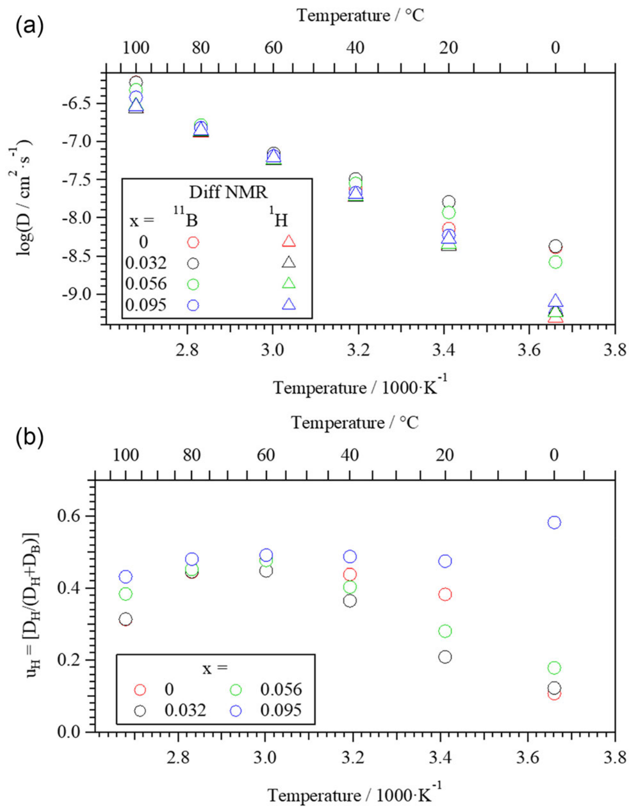

Figure 9. Self-diffusion coefficients for ^{1}H and ^{11}B on temperature obtained from PFG-NMR studies a). Diffusivity fraction (u_H) of protons on temperature calculated as $D_H/(D_H + D_B)$ b).

to pyrrolidinium cations. As the temperature is raised, cation–cation interactions within the ring stacks of pyrrolidinium cations become weaker, and Pyr_{14}^{+} ions are highly mobile, thus raising the diffusivity fraction. At temperatures above 60 °C, the intensity and frequency of the α-modes increase, leading to a coupling between the dynamics of the cations and anions, as previously shown for similar IL-based electrolytes.[12a] This diffusivity increase vs temperature depends more on the pyrrolidinium units than dynamics of boron complexes in anion domains. Consequently, the proton diffusivity fraction begins to decrease.

3. Conclusions

In this report, the ability of BCl_4^- anions to form stable $[BCl_4MgCl_2]^-$ dimer building blocks and to reveal the structure and interactions of proposed materials is exploited. The new electrolytes here prepared are based on 1-butyl-1-methylpyrrolidinium chloride IL and δ-$MgCl_2$. Eight $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes are obtained with x ranging from 0 to 0.095 (with $x = n_{Mg}/n_{IL}$). The elemental composition of the electrolytes is determined by ICP-AES analyses. The thermal stability and properties are studied by HR-TGA and MDSC measurements. Three different thermal events are detected (i.e., T_g, T_s, and T_m), which: 1) confirm that ions are organized in nanodomains; and 2) correspond to order-disorder transitions of these ion aggregates. It is found that doping the electrolytes with the magnesium salt significantly affects T_m. DFT computations: 1) allow to obtain the optimized geometries of the building blocks composing the electrolytes; and 2) demonstrate that δ-$MgCl_2$ units are coordinated by Lewis base ligands of the anion domains of the IL, forming $[BCl_4MgCl_2]^-$ species. FTIR investigations reveal the presence of three different concentration regions and point out that in electrolytes: 1) in region A, i.e., at low Mg concentration ($x < 0.032$), δ-$MgCl_2$ is coordinated preferentially by BCl_4^- Lewis base ligands of anionic domains, forming 3D catenated domains $[BCl_4\cdots MgCl_2]^-$ by bridging units; 2) in region B ($0.032 \le x \le 0.056$) the further added δ-$MgCl_2$ units are coordinated by the anionic species present in the matrix of catenated domains, thus raising the fraction of dynamic cross-links in anion domains of IL electrolytes; and 3) in region C, at high Mg concentrations ($x > 0.056$), the maximum solubility in electrolytes of δ-$MgCl_2$ is achieved. This suggests that in IL electrolytes, fragments of likely the typical "chain-like polymeric structure" of δ-$MgCl_2$ are present, which are coordinated by Lewis base sites of catenated anionic domains. BES studies: 1) show that $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_{0.041}$ sample exhibits the highest total ionic conductivity, with a value of 1.43×10^{-4} S cm^{-1} at 40 °C; 2) reveal that the electric response of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ is modulated by four dielectric events (α_1, α_2, β_1, and β_2) and four polarization phenomena (σ_{EP}, $\sigma_{IP,1}$, $\sigma_{IP,2}$, and $\sigma_{IP,3}$); 3) demonstrate that the detected phenomena and the dynamics of the IL nanostructure are correlated; and 4) allow to propose a reasonable conductivity mechanism, in which the diffusion of rotational states along the aggregation axis of Pyr_{14}^{+} pillars (α_1) predominantly plays a crucial role in assisting the long-range exchange of Mg species between different aggregates. Finally, NMR investigations allow us to demonstrate that chloroborate species are able to form a strong catenated coordination network, with a more electron-withdrawing effect with respect to the aluminum analogous IL-based electrolytes. In addition, PFG-NMR studies reveal that $[BCl_4]^-$ and $[BCl_4MgCl_2]^-$ species diffuse faster than $[Pyr_{14}]^+$ cations, thus demonstrating that species in anion nanodomains present a higher mobility with respect to that of cation aggregates.

In summary, this study demonstrates the crucial role of δ-$MgCl_2$ in modulating the structure and functions of proposed electrolytes. BES and NMR studies consistently show that Mg^{2+} transport phenomena, which are highly influenced by the temperature and magnesium concentration, occur through complex mechanisms involving electrode and interdomain polarizations (σ_{EP} and $\sigma_{IP,k}$). The overall conductivity mechanism is significantly affected by the α_1 and α_2 dielectric relaxation events of materials. δ-$MgCl_2$ promotes the formation of magnesium chloroborate complexes, increasing the disorder and enhancing the conductivity pathways of materials. The unique coordination network in IL electrolytes, established by doping with δ-$MgCl_2$, ensures in these materials more efficient long-range charge migration processes, even at elevated temperatures. Moreover, the observed high mobility of anionic species and the dynamic crosslinking in anion nanodomains suggest that these materials exhibit structural features that may facilitate Mg^{2+} transport, suggesting potential relevance for future studies on magnesium battery electrolytes.

These findings highlight two key breakthroughs in the design of IL electrolytes for magnesium batteries. First, the use of $[BCl_4]^-$ as both a structural building block and an in situ spectroscopic probe represents a novel approach for investigating and tailoring the nanoscale organization of electrolyte systems. Second, the formation of a dynamic, catenated coordination network mediated by δ-$MgCl_2$ introduces new opportunities to engineer mesoscale pathways for efficient Mg^{2+} transport.

While this work provides detailed structural and mechanistic insights into magnesium-ion transport in IL-based electrolytes, some limitations remain. In particular, the behavior of the proposed coordination networks in the presence of metallic magnesium electrodes and under electrochemical cycling conditions is yet to be determined. Future studies should address the stability of these nanostructures at electrode interfaces, their response to redox cycling, and the persistence of percolation pathways under dynamic conditions. These investigations will be crucial for translating the present findings into practical battery applications and for further optimizing the electrolyte formulation for interfacial and operational performance.

Taken all together, these insights are not only relevant for advancing the fundamental understanding of ionic conduction in complex IL systems, but also provide a foundation for the rational design of next-generation electrolytes with enhanced electrochemical performance and stability. As such, this work paves the way for the development of functional materials that show promise for further investigation as candidate electrolytes in high-energy magnesium-ion systems.

4. Experimental Section

Reagents: $Pyr_{14}Cl$ was supplied from io·li·tec. Metallic magnesium (50 mesh), 1.0 M boron trichloride solution in toluene, and 1-chlorobutane are Aldrich reagent grade. The IL was dried under vacuum at 105 °C for 7 days. 1-chlorobutane was further purified by standard methods and stored under Argon on 4 A molecular sieves to prevent moisture contamination. All transfer and handling operations were performed under a strictly argon atmosphere, either inside an argon-filled glove box or using a Schlenk line.

Synthesis: Anhydrous δ-$MgCl_2$ was obtained by direct reaction of metallic magnesium with 1-chlorobutane under a strictly anhydrous atmosphere, as described elsewhere.[21] The purity of δ-$MgCl_2$ was > 98 wt%. The $Pyr_{14}Cl/(BCl_3)_{0.25}$ IL was synthesized by reacting $Pyr_{14}Cl$ with a current flow of boron trichloride freshly distilled from a 1.0 M solution of boron trichloride in toluene. A schematic representation of the apparatus used for the preparation of the IL electrolytes is shown in Figure S1, Supporting Information. In detail, 100 mL of 1.0 M boron trichloride solution was inserted in a three-neck distillation flask connected to a receiving flask containing 12 g of $Pyr_{13}Cl$. The receiving flask was placed in an ice bath to keep the temperature down at 0 °C. The BCl_3 gas (b.p. 12.6 °C) was obtained by heating the BCl_3 solution in toluene (b.p. 110.6 °C) at 90 °C (Figure S1A, Supporting Information). The toluene vapors present in the gas flow were dropped back in the starting solution by a vertical reflux condenser set at 25 °C (Figure S1B, Supporting Information). By means of a distillation of 3 h, a sufficient amount of purified BCl_3 gas was obtained, which is then condensed in a Liebig condenser at −10 °C (Figure S1C, Supporting Information). The liquid boron trichloride thus obtained was then added drop by drop to the cold $Pyr_{14}Cl$, yielding the desired $Pyr_{14}Cl/(BCl_3)_{0.25}$ IL (Figure S1D, Supporting Information). The apparatus was assembled inside an Ar-filled glove box. A strict argon atmosphere obtained from a Schlenk line was maintained in the reaction apparatus by fluxing Ar from the distilling flask. The synthesized IL was collected, dried at 90 °C for 6 h in order to eliminate possible toluene traces, and stored inside a glove box. The most concentrated ($[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_{0.095}$) electrolyte was obtained by preparing a saturated solution of δ-$MgCl_2$ in the pure IL. By dilution of this electrolyte with pristine $Pyr_{14}Cl(BCl_3)_{0.25}$ IL, a family of $[Pyr_{14}Cl/(BCl_3)_{0.25}]/(\delta\text{-}MgCl_2)_x$ electrolytes with x ranging from 0 to 0.095 was obtained. The composition of the electrolytes determined by ICP-AES measurements is summarized in Table 1.

Composition and Thermal Properties: The chemical composition of the proposed electrolytes was determined by inductively coupled plasma atomic emission spectroscopy analyses (ICP SPECTRO Arcos EOP). HR-TGA measurements were carried out using a high-resolution thermobalance TGA2950 (TA Instruments). A constant flow of 100 sccm of dry nitrogen during the measurements was adopted. The instrumental HR-TGA sensitivity ranges from 0.1 to 2% min^{-1}, and its resolution was 1 μg. HR-TGA profiles were collected from 25 to 950 °C. The heating rate was modulated on the basis of the first derivative of the weight loss and ranges from 40 to 0.001 °C min^{-1}. MDSC investigations were performed on a Q20 instrument from TA Instruments. The heating rate is 3 °C min^{-1} and the temperature range is between −100 and 100 °C. An amount of ≈ 8 mg was inserted in a hermetically sealed aluminum pan inside an Argon dry-box.

Computational Studies: DFT studies were carried out using DMol3 package with a GGA basis set and BLYP functionals with the computational software Materials Studio 2016.

Structure and Interactions: FT-FIR studies were performed in transmission using a Nicolet FTIR Nexus spectrometer in the range from 600 to 50 cm^{-1}. Samples were sandwiched between two hermetically sealed polyethylene windows inside dry box. ATR-FT-MIR spectra were acquired using a Nicolet iS50 instrument equipped with a Golden Gate diamond single beam Attenuated Total Reflectance cell.

Electrical Properties: The electric response of the IL-based electrolytes was investigated by BES technique. Permittivity spectra were collected in the frequency range from 30 mHz to 10 MHz, and from −100 to 100 °C with a step of 10 °C and an accuracy greater than ±0.1 °C using a Novocontrol Alpha-A Analyzer. The sample temperature was controlled using a homemade thermostat operating with liquid nitrogen from −190 to 300 °C. The samples sandwiched between two platinum electrodes of 13 mm diameter were inserted inside a sealed homemade Teflon cell. In order to maintain constant the thickness of the samples and to prevent any short circuit during measurements, an optical fiber with a diameter of $d = 0.125$ mm was used as spacer between the two electrodes.

NMR Studies: 1D ^1H spectral and ^1H and ^{11}B PFG-NMR analyses were performed on a 7.05 T (^1H and ^{11}B Larmor frequencies of 300.13 and 96.29 MHz, respectively) Varian-S Direct Drive Wide Bore spectrometer equipped with a z-gradient probe (DOTY Scientific, Inc.). 1D ^1H NMR spectra were acquired at 20 °C using tetramethylsilane (TMS) as a reference at 0 ppm. In PFG-NMR analyses, using a stimulated-echo pulse sequence, sixteen field gradient values were linearly increased from 5 to 350 or 810 G cm^{-1} for ^1H or ^{11}B, respectively. The ^1H or ^{11}B signal of each increment was collected over 16 or 64 transients with 3.5 or 2 s recycling delay, respectively. Field gradient pulse durations (δ) of 3 ms (^1H) or 2 ms (^{11}B) and diffusion delays (Δ) of 0.6 s (^1H) or 0.2 s (^{11}B) are used. 1D ^{11}B NMR studies were carried out using a 400 MHz (^{11}B Larmor frequency of 128.37 MHz) Bruker Avance III instrument. Measurements are carried out at 20 °C using boric acid as reference at 0 ppm. ^{25}Mg NMR spectra were acquired on a Bruker AVANCE NEO 750 MHz (17.62 T) wide bore UltraStabilized spectrometer operating at 45.91 MHz. A Hahn-echo nonrotor-synchronized pulse sequence with a short echo time (80 μs) was used in order to reduce the background noise. 90° and 180° pulse-lengths of 8 and 16 μs, respectively, were used for ^{25}Mg NMR studies with a recycle delay of 0.1 s and a number of transients comprised between 10'000 and 200'000 (depending on magnesium concentration and temperature). Samples were hermetically sealed under Ar using Kel-F drive tips in 4 mm zirconia rotors. An 11 M $MgCl_2$ aqueous solution was used as an external reference set at 0 ppm. The temperature was varied, ranging between 0 and 80 °C, collecting spectra every 20 °C. The temperature was calibrated by monitoring the chemical shift variation of a lead nitrate standard.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Vito Di Noto, **Steven G. Greenbaum**, and **Gioele Pagot** conceived the idea for the project and planned the experiments. **Gioele Pagot** synthesized the electrolytes and performed the thermal analyses and the vibrational spectroscopy investigations. **Keti Vezzù** completed and analyzed the BES studies. **Mounesha N. Garaga** accomplished the 1D and PFG ^{1}H and ^{11}B NMR investigations. **Boris Itin** performed the 1D-^{25}Mg-NMR studies. All the authors discussed the results, conceived the conclusions, and contributed to the final version of the manuscript. **Gioele Pagot** drafted the manuscript with the support and supervision of **Vito Di Noto** and **Steven G. Greenbaum**.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

broadband electrical spectroscopy, chloroborate coordination network, ionic liquid electrolytes, magnesium batteries, nuclear magnetic resonance

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